Pasan Ariyarathna

Software Engineering | AI/ML | Full-Stack Development
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Professional Summary

Third-year BSE (Hons) Information Technology undergraduate with practical experience designing and developing software projects across AI/ML and full-stack development. Skilled in Python, JavaScript/TypeScript, backend APIs, database integration, and modern web technologies. Experienced in building LLM-powered applications, multi-agent workflows, machine-learning pipelines, and microservices through academic and team projects. Enjoys solving problems, learning new technologies, collaborating with others, and turning ideas into reliable, user-focused software.

Technical Skills

Programming: Python, JavaScript, TypeScript, Java, Kotlin, SQL, C
AI/ML: LLM APIs (Gemini, OpenAI, Groq/Llama), multi-agent systems, RAG, embeddings, FAISS, HuggingFace, scikit-learn, SHAP, Particle Swarm Optimization
Web Development: React, Next.js, Streamlit, Vite, HTML, CSS, Tailwind CSS
Backend & APIs: FastAPI, Spring Boot, REST APIs, JWT authentication, API-key authentication
Databases: PostgreSQL, MongoDB, MySQL, Supabase, Firebase/Firestore, Redis
Engineering Tools: Git/GitHub, Docker, Docker Compose, GitHub Actions (CI/CD), Linux, unit and integration testing

Selected Projects

Multiagent Tutor *Python, Streamlit, Groq/Llama, FAISS, HuggingFace, RAG*
GitHub: github.com/PasanAriyarathna/multiagent-tutor

- Built an AI-powered Python tutoring application with six specialised agents: Router, Explainer, Quiz, Evaluation, Remedial, and Report.
- Implemented Retrieval-Augmented Generation (RAG) using FAISS and HuggingFace embeddings to ground responses in PDF lesson content.
- Added code-aware answer evaluation and remedial practice tailored to learners' weak concepts.

Explainable Heart-Disease Risk Predictor *Python, scikit-learn, PSO, SHAP, FastAPI, Groq/Llama*
GitHub: GitHub Repository

- Developed a machine-learning pipeline using a 50,000-record dataset, covering data preprocessing, model training, and evaluation.
- Applied binary Particle Swarm Optimization to select seven features from 20 clinical features.
- Compared Logistic Regression, Naive Bayes, K-Nearest Neighbours, and Random Forest; the reported Logistic Regression test F1-score was 0.917.
- Exposed predictions and SHAP feature contributions through FastAPI and integrated Groq/Llama to provide plain-language explanations.

CeyGreen – Greenhouse Management Platform (Team Project) *Spring Boot, React, PostgreSQL, Kafka, Docker, Azure*
GitHub: github.com/vikumkodikara/CeyGreen

- Contributed to a greenhouse management platform with microservices and a treatment-recommendation service mapping plant disease diagnoses to suitable remedies.
- Developed REST API functionality and PostgreSQL-backed remedy data, including database migrations, search, alternative remedies, and farmer ratings.
- Integrated Kafka events and API security using API-key filtering and JWT role checks.
- Worked with Docker Compose and GitHub Actions in the project's build and deployment workflow; cloud platform: Microsoft Azure.

Education

BSc (Hons) in Information Technology *2023 – Present*
Horizon Campus, Sri Lanka

Work Experience

Non-Academic Staff Trainee – Examination Division
University of Vocational Technology, Rathmalana *Sep 2022 – Mar 2023*

- Supported daily examination-division operations, coordinating with colleagues and completing tasks accurately within deadlines.
- Developed communication, teamwork, time-management, and problem-solving skills in a professional environment.

Competitions & Achievements

Cursor Buildathon – Gemini Track
Placed 10th among participating teams with an AI-driven solution.

AI Challenge Sri Lanka 2026 – AI Buildathon
Participated as a member of Syntax Squad, contributing to an AI-driven project focused on solving real-world challenges.

Reference

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